

DEPARTMENT OF COMPUTER SCIENCE

ASSIGNMENT

Descriptive And Frequentist Statistics

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Descriptive Statistics is the process of summarizing and describing the main features of a dataset.

- Organizes data
- Summarizes data
- Describes the distribution of data
- Measures central tendency and spread
- Helps in understanding large datasets

== Topics Covered

- 1 Histogram
- 2 Sample Mean
- 3 Sample Variance
- 4 Order Statistics
- 5 Sample Covariance
- 6 Sample Covariance Matrix

Histogram

A **histogram** is a graphical representation of the distribution of numerical data.

- Data is divided into intervals called bins.
- The height of each bar represents the frequency.
- It helps identify the shape and spread of data.

Example:

Consider the data:

2, 3, 3, 4, 5, 5, 5, 6, 7, 8

The data can be divided into intervals and represented using a histogram.

Sample Mean

The **sample mean** is the average of all observations in a sample.
If the observations are

$$x_1, x_2, \dots, x_n$$

then the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example:

For the data

$$2, 4, 6, 8, 10$$

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Therefore,

Sample Variance

Sample variance measures how far the observations are spread around the sample mean.

The formula is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where:

- s^2 = sample variance
- n = sample size
- $\bar{x}$ = sample mean

Sample Variance – Example

Consider:

$$2, 4, 6, 8, 10$$

The sample mean is

$$\bar{x} = 6$$

Therefore,

$$\begin{aligned}s^2 &= \frac{(2 - 6)^2 + (4 - 6)^2 + (6 - 6)^2 + (8 - 6)^2 + (10 - 6)^2}{5 - 1} \\ &= \frac{16 + 4 + 0 + 4 + 16}{4}\end{aligned}$$

$$\boxed{s^2 = 10}$$

Understanding Variance

Variance measures the average squared distance of observations from their mean.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Small variance $\rightarrow$ observations are close together.
- Large variance $\rightarrow$ observations are widely spread.

Order Statistics

Order statistics are the observations arranged in increasing order.
Suppose the sample is

$$8, 3, 6, 2, 9$$

After arranging:

$$2, 3, 6, 8, 9$$

The order statistics are written as

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}$$

where

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

Order Statistics – Example

Consider:

12, 5, 8, 3, 10

Ordered data:

3, 5, 8, 10, 12

Therefore,

$$X_{(1)} = 3$$

$$X_{(2)} = 5$$

$$X_{(3)} = 8$$

$$X_{(4)} = 10$$

Sample Covariance

Sample covariance measures the relationship between two variables.
For variables X and Y :

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

where:

- $\bar{x}$ = mean of X
- $\bar{y}$ = mean of Y
- n = number of observations

Interpretation of Covariance

The sign of covariance describes the direction of the relationship.

- $s_{XY} > 0$: Positive relationship
- $s_{XY} < 0$: Negative relationship
- $s_{XY} \approx 0$: Little or no linear relationship

Example:

If study time increases and examination marks also increase, the covariance is generally positive.

Sample Covariance – Example

Consider:

$$X = (1, 2, 3)$$

$$Y = (2, 4, 6)$$

Means:

$$\bar{x} = 2, \quad \bar{y} = 4$$

Therefore,

$$\begin{aligned}s_{XY} &= \frac{(1 - 2)(2 - 4) + (2 - 2)(4 - 4) + (3 - 2)(6 - 4)}{3 - 1} \\ &= \frac{2 + 0 + 2}{2}\end{aligned}$$

$$s_{XY} = 2$$

Sample Covariance Matrix

For multiple variables, covariances can be arranged in a matrix.
For two variables X and Y :

$$S = \begin{bmatrix} s_X^2 & s_{XY} \\ s_{YX} & s_Y^2 \end{bmatrix}$$

Since

$$s_{XY} = s_{YX},$$

the covariance matrix is symmetric.

Covariance Matrix – Example

Suppose:

$$s_X^2 = 4, \quad s_Y^2 = 9, \quad s_{XY} = 2$$

Then the covariance matrix is

$$S = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$$

Diagonal elements represent variances.

Off-diagonal elements represent covariances.

General Covariance Matrix

For p variables, the sample covariance matrix is

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

where

$$s_{ij} = \frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

Measures of Dispersion

Dispersion describes how widely the observations are spread around the center.

Important measures:

- Range
- Variance
- Standard deviation
- Interquartile range

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example:

10, 12, 15, 18, 20

$$\text{Range} = 20 - 10 = 10$$

Interquartile Range (IQR)

The Interquartile Range measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

where:

- Q_1 = First quartile
- Q_3 = Third quartile

Example:

$$Q_1 = 15, \quad Q_3 = 25$$

Therefore:

$$IQR = 25 - 15 = 10$$

IQR is relatively resistant to extreme observations.

Box Plot

A box plot summarizes data using five important values:

- 1 Minimum
- 2 First quartile (Q_1)
- 3 Median
- 4 Third quartile (Q_3)
- 5 Maximum



The box represents the **IQR** and the lines extending from the box are called **whiskers**.

Outliers

An outlier is an observation that is unusually far from the remaining observations.
Using the IQR method:

$$\text{Lower Fence} = Q_1 - 1.5(IQR)$$

$$\text{Upper Fence} = Q_3 + 1.5(IQR)$$

Observations below the lower fence or above the upper fence are commonly classified as outliers.

Example:

$$Q_1 = 10, \quad Q_3 = 20$$

$$IQR = 10$$

Histogram

A histogram represents the distribution of numerical data.

The observations are divided into intervals called **bins**, and the height of each bar represents the frequency.

Bin	Frequency
0–10	3
10–20	7
20–30	5
30–40	2

A histogram helps identify:

- Center
- Spread
- Distribution shape
- Skewness
- Possible outliers
- Number of modes

Skewness

Skewness measures the **asymmetry** of a distribution.

Symmetric

$$\text{Mean} \approx \text{Median}$$

Positive / Right Skew

The distribution has a longer right tail.

Typically:

$$\text{Mean} > \text{Median}$$

Negative / Left Skew

The distribution has a longer left tail.

Typically:

$$\text{Mean} < \text{Median}$$

Kurtosis

Kurtosis describes aspects of the **tail heaviness** of a distribution.
One population definition is:

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

For a normal distribution:

$$\text{Kurtosis} = 3$$

Excess kurtosis is:

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3$$

Therefore, normal distribution has excess kurtosis:

$$0$$

Concept	Purpose
Histogram	Shows data distribution
Sample Mean	Measures central value
Sample Variance	Measures data spread
Order Statistics	Arranges observations
Covariance	Measures relationship
Covariance Matrix	Summarizes multiple covariances

Conclusion

Descriptive statistics provides useful methods for summarizing and understanding data.

- Histograms show the distribution of data.
- Mean describes the center of a sample.
- Variance describes the spread.
- Order statistics arrange observations.
- Covariance describes relationships between variables.
- Covariance matrices summarize relationships among several variables.

Frequentist Statistics is a statistical approach where parameters are considered fixed but unknown, while data are treated as random.

Main topics:

- Sampling
- Mean Square Error
- Consistency
- Confidence Intervals
- Parametric Model Estimation
- Non-Parametric Model Estimation

1. Sampling

Definition:

Sampling is the process of selecting a smaller group (sample) from a larger population to collect information about the population.

Example:

Suppose a college has 1000 students. We randomly select 100 students to study their average height.

Population = 1000 students

Sample = 100 students

The sample is used to estimate the characteristics of the population.

Sampling Example

Suppose the heights of 5 selected students are:

160, 165, 170, 175, 180

The sample mean is:

$$\bar{x} = \frac{160 + 165 + 170 + 175 + 180}{5}$$

$$\bar{x} = 170 \text{ cm}$$

Thus, the estimated average height of the population is approximately 170 cm.

2. Mean Square Error (MSE)

Definition:

Mean Square Error measures how far an estimator is from the true parameter value.

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

It can also be written as:

$$MSE(\hat{\theta}) = \text{Variance}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

where

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

MSE Example

Suppose the true value is:

$$\theta = 10$$

and an estimator gives:

$$\hat{\theta} = 12$$

Then:

$$MSE = (12 - 10)^2$$

$$MSE = 2^2$$

$$MSE = 4$$

A smaller MSE indicates that the estimator is generally closer to the true parameter.

3. Consistency

Definition:

An estimator is called **consistent** if it gets closer to the true parameter as the sample size increases.

Mathematically:

$$\hat{\theta}_n \xrightarrow{P} \theta \text{ as } n \rightarrow \infty$$

Here, $\xrightarrow{P}$ means convergence in probability.

Consistency Example

Suppose the true population mean is:

$$\mu = 50$$

As the sample size increases, the sample mean may behave like:

n	$\bar{x}$
10	52
50	50.8
100	50.3
500	50.1
1000	50.02

As n increases:

$$\bar{x} \rightarrow 50$$

Therefore, the sample mean is a **consistent estimator** of the population mean.

4. Confidence Intervals

Definition:

A confidence interval gives a range of values that is used to estimate an unknown population parameter.

For a population mean with known standard deviation:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

For a 95% confidence interval:

$$z_{\alpha/2} = 1.96$$

Confidence Interval Example

Suppose:

$$\bar{x} = 50, \quad \sigma = 10, \quad n = 100$$

For a 95% confidence interval:

$$\begin{aligned} CI &= 50 \pm 1.96 \frac{10}{\sqrt{100}} \\ &= 50 \pm 1.96 \end{aligned}$$

Therefore:

$$CI = (48.04, 51.96)$$

Thus, the 95% confidence interval for the population mean is approximately 48.04 to 51.96.

5. Parametric Model Estimation

Definition:

Parametric estimation assumes that the population follows a specific probability distribution with a fixed number of parameters.

For example:

$$X \sim N(\mu, \sigma^2)$$

The parameters are:

$$\theta = (\mu, \sigma^2)$$

A common method of estimation is **Maximum Likelihood Estimation (MLE)**:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Parametric Estimation Example

Suppose the observations are:

$$8, \quad 10, \quad 12, \quad 10, \quad 10$$

Assume:

$$X \sim N(\mu, \sigma^2)$$

Estimate μ using the sample mean:

$$\hat{\mu} = \frac{8 + 10 + 12 + 10 + 10}{5}$$

$$\boxed{\hat{\mu} = 10}$$

Therefore, the estimated population mean is 10.

6. Non-Parametric Model Estimation

Definition:

Non-parametric estimation does not assume a specific probability distribution for the population.

It is useful when the distribution of the data is unknown.

Common methods include:

- Empirical distribution
- Kernel density estimation
- Rank-based methods
- Median estimation

Non-Parametric Example

Suppose the data are:

5, 7, 8, 10, 12, 15

The median is:

$$\text{Median} = \frac{8 + 10}{2}$$

$$\boxed{\text{Median} = 9}$$

Here, we estimate the center of the data using the median without assuming that the data follow a normal distribution.

Thus, this is an example of a non-parametric approach.

Parametric vs Non-Parametric

	Parametric	Non-Parametric
Distribution assumption	Required	Not required
Parameters	Fixed number	Not fixed
Example	Normal distribution	Median
Flexibility	Lower	Higher
Data requirement	More assumptions	Fewer assumptions

- **Sampling:** Selects a sample from a population.
- **MSE:** Measures the error of an estimator.
- **Consistency:** An estimator approaches the true parameter as sample size increases.
- **Confidence Interval:** Gives a range for estimating an unknown population parameter.
- **Parametric Estimation:** Assumes a specific probability distribution.
- **Non-Parametric Estimation:** Requires fewer distributional assumptions.

Conclusion

Frequentist statistics provides methods for making conclusions about populations using sample data.

The important concepts are:

Sampling → Estimation → Confidence → Inference